Denis Gruszkiewicz

Selected ML Research & AI Projects

r3dflzp.ns@gmail.com | github.com/Flyingzaptop | Wrocław, Poland

RESEARCH APPROACH

My independent research has focused on financial time series. The transferable problems are broader: building reliable datasets, defining meaningful targets, preventing leakage, comparing architectures and understanding why a model fails. I use explicit baselines and causal tests, and document limitations alongside results.

1. XAUUSD: NEURAL SEQUENCE EXPERIMENTS

Explored angular representations, causal temporal convolutional networks and asynchronous event inputs with attention pooling in PyTorch. Separate experiments investigate forecasting and bidirectional reconstruction.

- Designed custom losses and output constraints; compared reconstruction approaches against interpolation baselines.
- Added causality checks to prevent future events entering forecasting inputs. Reconstruction results did not establish a clear advantage over the strong baseline.

https://github.com/Flyingzaptop/xauusd

2. STARGAZIN: MODEL BEHAVIOR AND PIPELINE CONSISTENCY

Built a pipeline from Parquet ingestion and normalization through features, labels, training/checkpoints and offline scoring. Explored CT-LSTM and RetNet-lite variants, output-head redesigns and margin objectives.

- Investigated collapsed or unstable predictions and consistency between offline and online preprocessing.
- Used Rust/PyO3 extensions for selected preprocessing paths, with unit, integration and regression tests.

https://github.com/Flyingzaptop/stargazin

3. GC MBO RESEARCH: EVALUATION UNDER REALISTIC ASSUMPTIONS

Investigated event-level gold-futures data using Python feature engineering and ML experiments, Rust replay and conservative FIFO execution simulation.

- Used walk-forward evaluation, causality checks and latency/cost stress to distinguish model signals from artifacts of the evaluation setup.
- The published research did not establish a stable executable trading edge. Its value lies in reusable infrastructure, explicit assumptions and documented rejected hypotheses.

https://github.com/Flyingzaptop/gc-mbo-research

FROM RESEARCH TO APPLICATIONS

JobHunter combines local LLM analysis, deterministic validation, n8n/Node.js workflows, PostgreSQL and human review. LabFlow is a Django/React demonstrator with RBAC, audit history, a local Ollama streaming client and optional faster-whisper speech-to-text.

https://github.com/Flyingzaptop/jobhunter-n8n | https://github.com/Flyingzaptop/LabFlow

CONTEXT

These are independent and academic projects, alongside my work as a Junior IT Administrator. I am studying Robotics and Process Automation (five semesters completed) and IT systems/network engineering. Code, methodology and documented results are available in the repositories; full reproduction can require local datasets, model checkpoints or suitable hardware.